



Evolving the CMS Submission Infrastructure

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HTC26

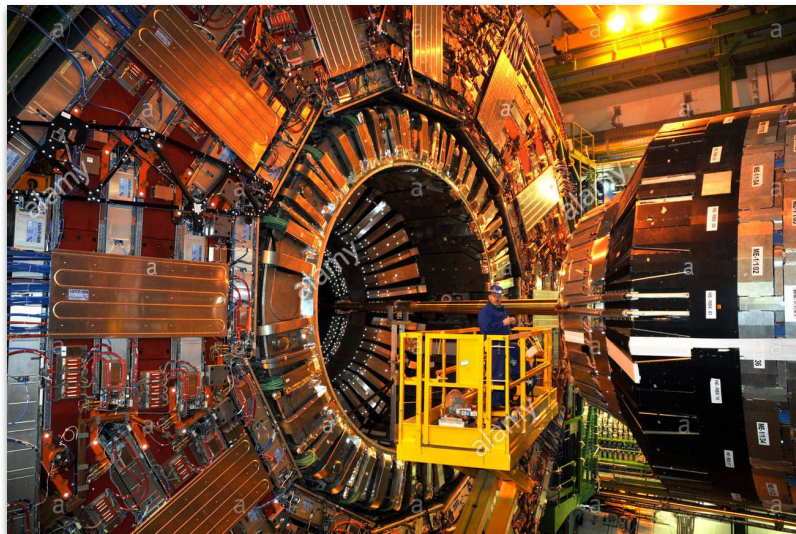


Outline

- CMS computing at scale
 - CERN, CMS and the WLCG ecosystem
- The CMS Submission Infrastructure
 - Supporting Tier-0, production and user analysis
- Improving resource utilization
 - Group surplus, IO-slots and pilot overloading
- Towards HL-LHC-scale workflow management

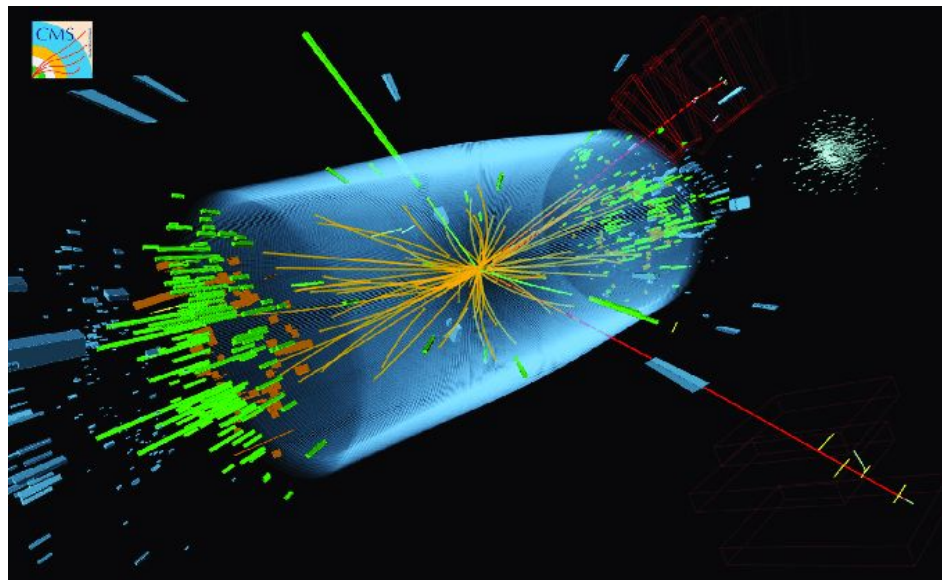


The CMS experiment at CERN



- Experimental data is reconstructed, distributed and analyzed together with large-scale Monte Carlo simulations.
 - **Hundreds of PB of data processed every year**

- One of the four major experiments at the **LHC at CERN**, where **proton-proton interactions** are produced at the world's **highest collision energy**
- **Studies elementary particles and fundamental forces in a broad range of physics processes**

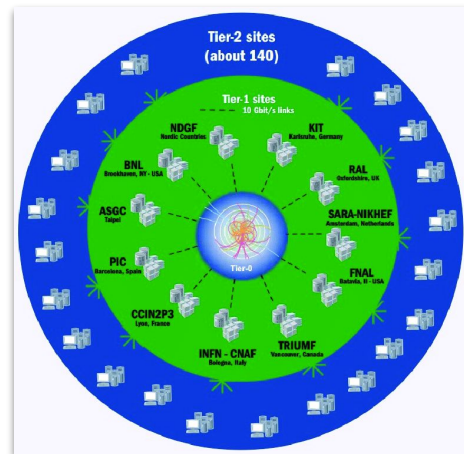




The Worldwide LHC Computing Grid



- CMS data is processed and analyzed using the **Worldwide LHC Computing Grid (WLCG)**
 - ~170 computing centers distributed worldwide
 - ~1.4 million CPU cores and ~2 EB of storage
 - Resources organized in Tier-0, Tier-1 and Tier-2 facilities

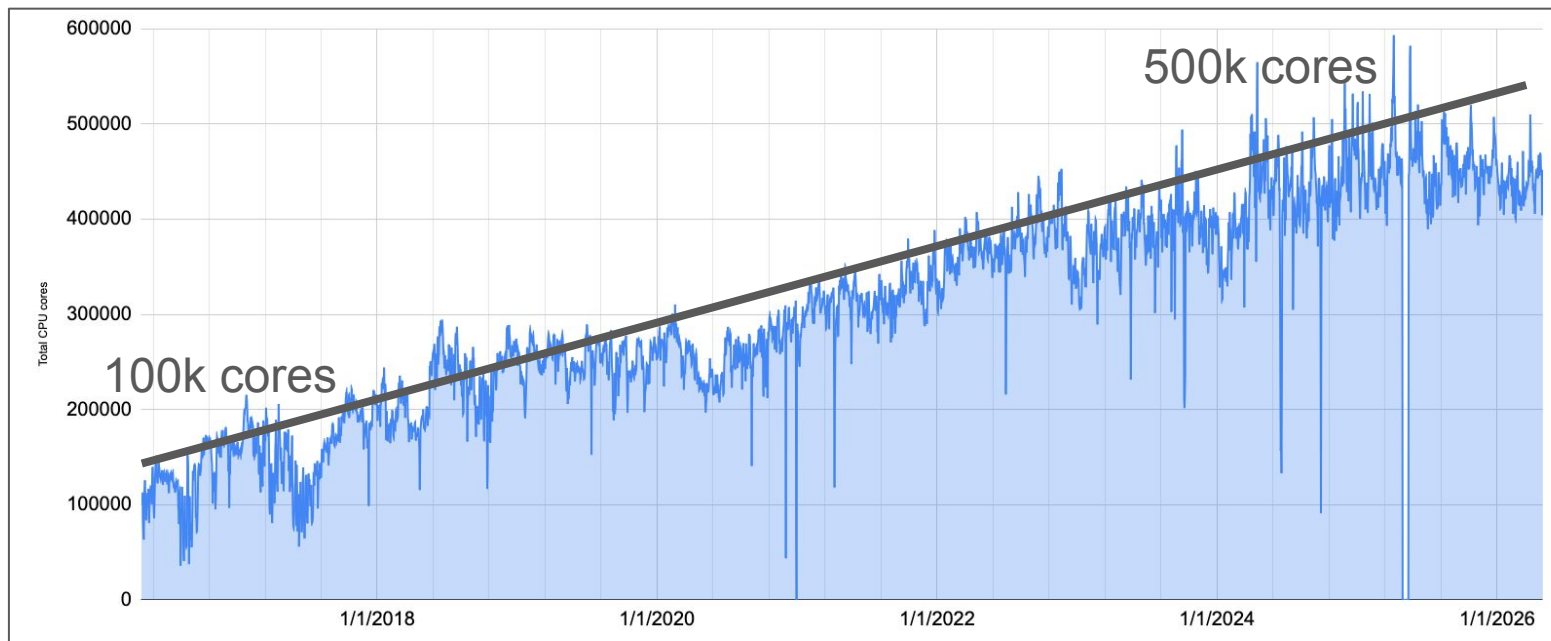




Evolution of total CPU capacity managed by **CMS** with HTCondor over the years



- Continuous growth from ~100k to ~500k cores over the last decade
- HTCondor provides a single federated pool spanning WLCG, HPC and cloud resources



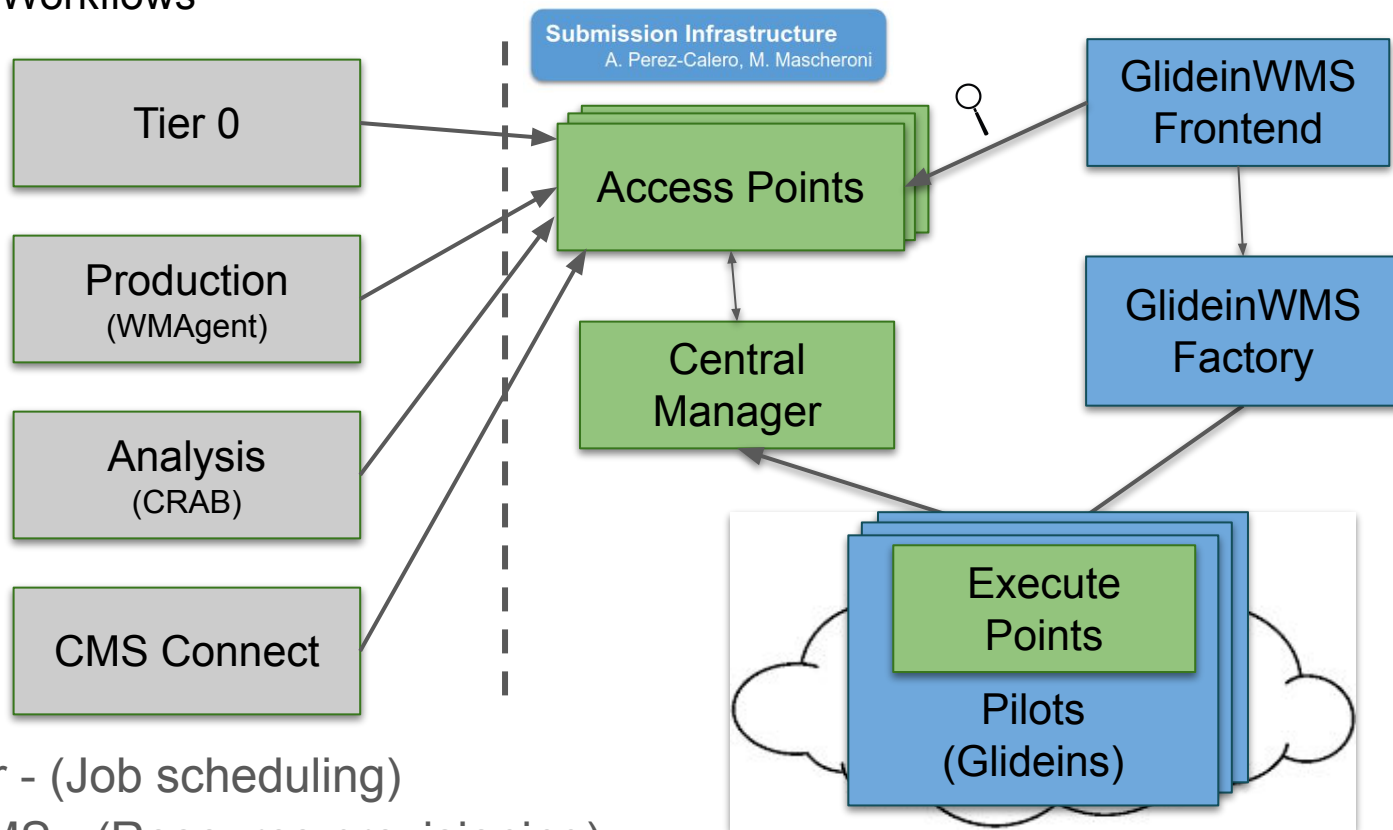


Architecture of the CMS Submission Infrastructure



Workflows

Jobs (SI)





Resource provisioning and partitioning in the CMS SI



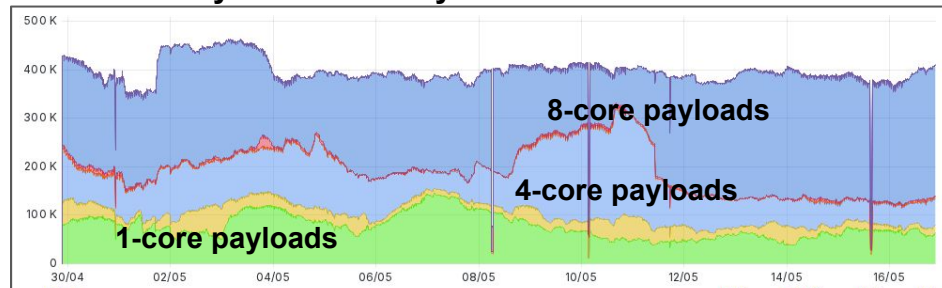
Acquiring resources:

- Mostly **16-core** and **8-core** pilots.
- Also **whole-node** and **HPC** resources.
- No need for dedicated **high-memory** slots, average ~2.5 GB/core

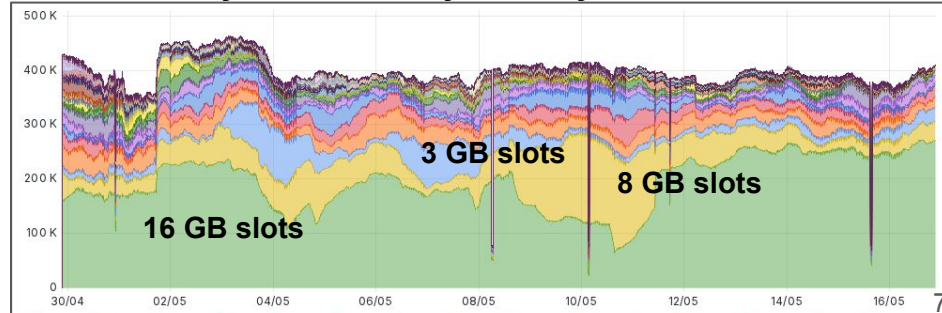
Using resources:

- Dynamic partitioning by CPU and memory
- 1-, 4- and 8-core payloads are the most common
- Production and analysis, and Tier 0 coexist in the same pool

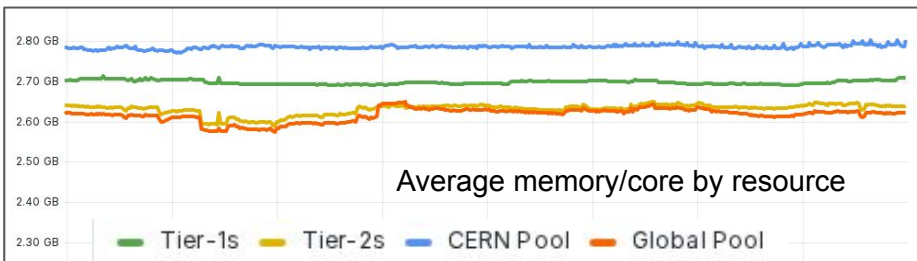
Dynamic slots by CPU cores



Dynamic slots by memory



Average memory/core by resource



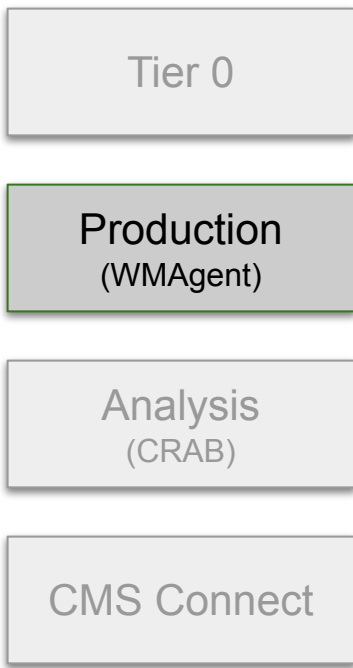
Tier 0

Production
(WMAgent)

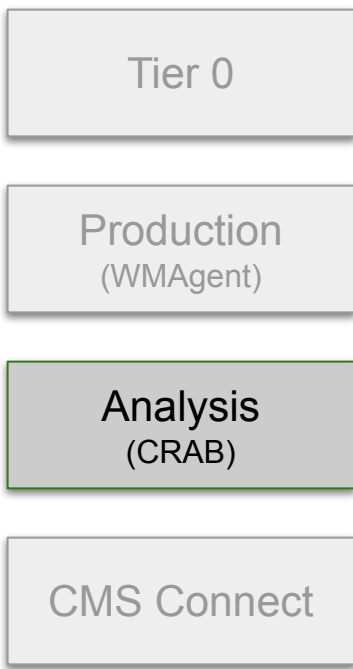
Analysis
(CRAB)

CMS Connect

- **RAW event** data must be formatted and reconstructed
 - **Repack** jobs (single-core, highest priority)
 - **Prompt Reconstruction** jobs (multicore)
- **Latency-critical**
 - **~1 PB/day** during data taking
 - Delays risk filling online buffers



- Large-scale campaign processing
 - **Monte Carlo production** campaigns
 - Reprocessing and re-reconstruction of **data**
- Throughput-oriented
 - **Creation of** MiniAOD and NanoAOD **datasets**
 - Flexible workloads **optimized** for high resource utilization



- **User-driven** workflows
 - Private Monte Carlo productions
 - Detector studies
 - Final physics analyses
- **Highly diverse**
 - Histograms from NanoAOD
 - Ntuplizers, skims and custom workflows
 - Many specialized use cases
 - Mostly **single-core**



Sharing resources across Tier0, Production and Analysis



- Priority resources
 - Most **CERN** resources are reserved for **Tier0 workloads**
- Flexible sharing
 - Production and analysis can **temporarily exceed their quota**
 - Enabled through `GROUP_ACCEPT_SURPLUS`
 - During LHC **shutdown periods**, resources naturally evolve toward a **production–analysis balance** (e.g.: 50/50 split at CERN)

Group	Tier0	Production	Analysis
CERN	80%	10%	10%
T1	30%	63%	7%
US T2	10%	63%	27%
non-US T2 + T3	0%	60%	40%

Divisions in group possible thanks to **pool federation** and job's **flocking**



Single-core bursts create fragmentation and Tier-0 starvation

Group surplus disabled



- Fragmentation

- Bursty **single-core repack jobs** create internal fragmentation
- Single-core **analysis** jobs opportunistically **fill the gaps**

- Consequences

- **PromptReco multicore jobs** may temporarily **starve**
- Disabling `GROUP_ACCEPT_SURPLUS` restores Tier0 quotas
- Small scheduling inefficiencies introduced by repack jobs retraction once analysis cannot fill up the gaps

Tradeoff: efficiency versus Tier0 latency





Dedicated IO-slots reduce fragmentation caused by auxiliary jobs

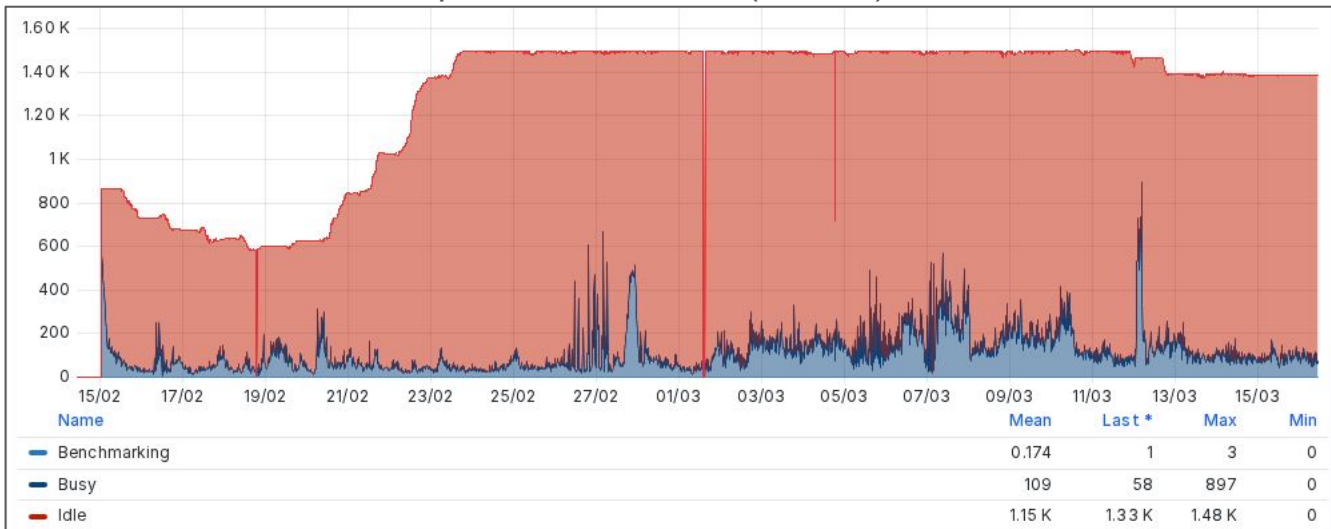
Fragmentation

- Short single-core auxiliary jobs (merge, log collect, ...) arrive in bursts
- Internal fragmentation impacts prioritization and efficiency
- High-priority multicore jobs may not find large slots

Dedicated IO-slots

- CMS SI introduces an additional **IO-slot** per pilot
- Reserved exclusively for auxiliary payloads
- The slot is **virtually added** and often idle
- Large slots are preserved and fragmentation is reduced

Example IO-slots in use (Tier-1s)



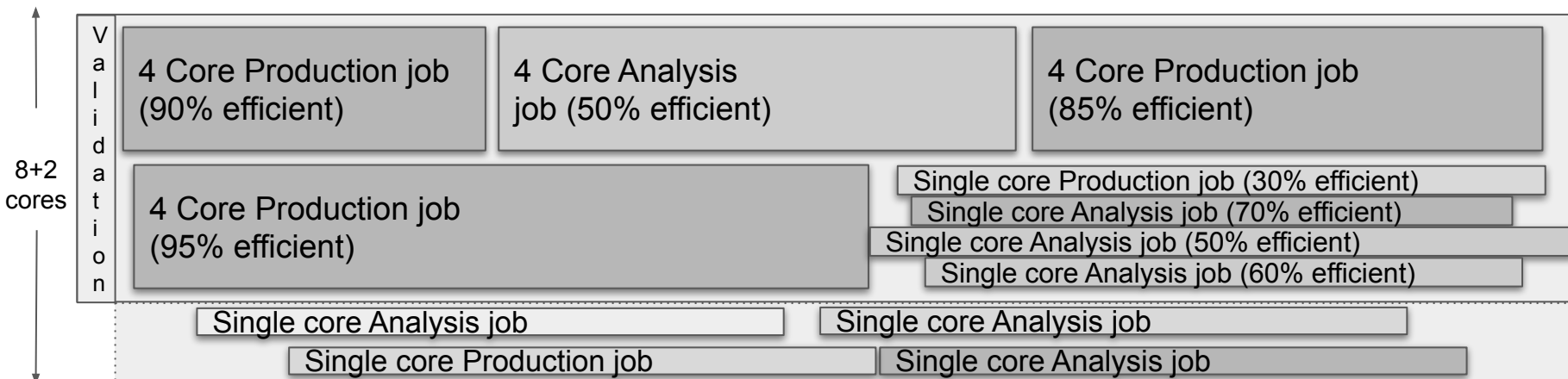
Recover unused CPU cycles: overloading pilots

Principle

- Recover **unused CPU cycles**, not new resources
- Push more workload into the same pilot envelope
- Moderate overloading (+25% CPU and memory)
- Additional payload jobs match extra **virtual cores**

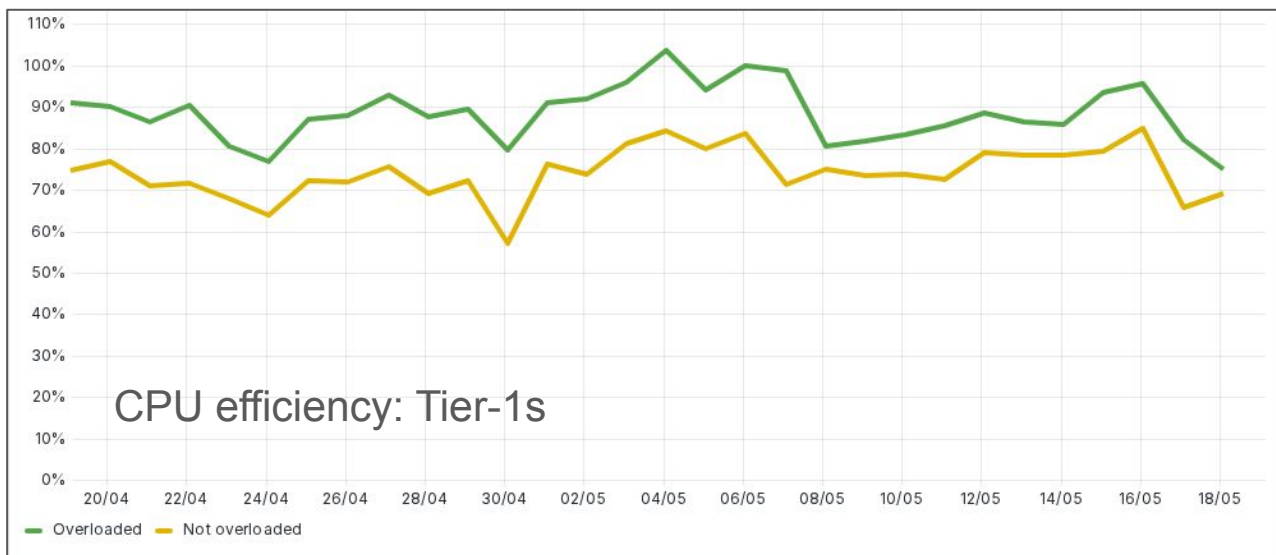
Benefits

- Improve overall CPU utilization
- No workflow changes required
- Implemented entirely from the SI side
- Easily deployed and tested



Overloading results

- **Introduced in 2023** as a test and progressively extended
 - **Subsequently** deployed on about $\frac{2}{3}$ of the **suitable resources**
 - Critical **CERN resources excluded** due to the Tier-0 tasks they host
- Pilot **CPU utilization measured** as total Sys+User CPU time relative to Walltime $\times$ Ncores
 - Allows computation of the full pilot CPU efficiency



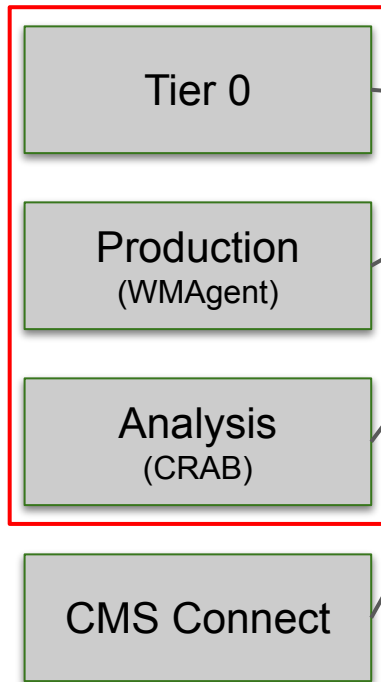
Significant and systematic improvement observed



Towards a sustainable Workflow Management system for HL-LHC



Workflows



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Access Points

Central Manager

Jobs (SI)

GlideinWMS
Frontend

GlideinWMS
Factory

Execute
Points

Glideins

Funding and available manpower make **convergence** towards a **common workflow solution** increasingly important.

- [WM Review](#) in 24/25 concluded that the current system is not viable for HL-LHC
 - a. Recommended to consider PanDA, DIRAC-X, or revamping the current system
- Several round of investigations with various group and people involved
- Consensus emerged during Spring 2026 Offline and Computing week

DIRAC-X



- Community solution
- Preferred direction

Revamped WM



- Micro-agent architecture
- Fallback solution

PanDA



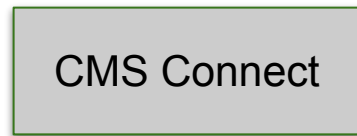
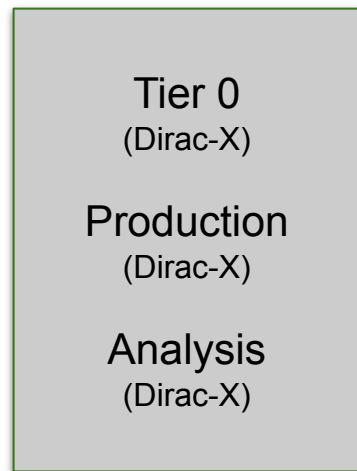
- Evaluated during review
- Not actively pursued



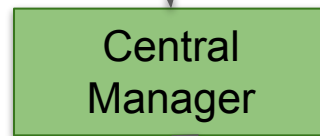
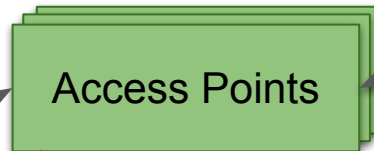
The Submission Infrastructure as a foundation for HL-LHC



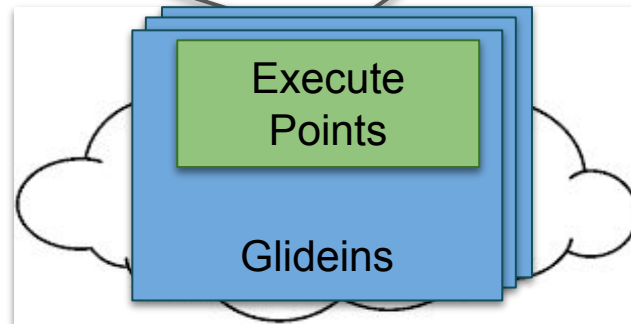
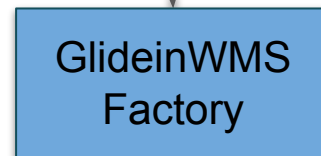
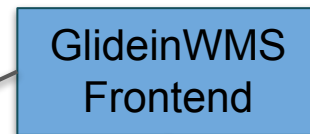
Workflows



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Jobs (SI)



Plan to use Dirac-X transformation system together with current glideinWMS/Condor infrastructure



Conclusions

- CMS Submission Infrastructure continuously **evolved for the past 10 years**
 - Integrating increasingly **heterogeneous resources**
 - Improving resource utilization through **IO-slots** and **pilot overloading**
 - Supporting diverse workloads within a **common** federated **HTCondor pools**
- **Proven at scale**
 - Grew from **100k to 500k cores**
 - Federates WLCG, HPC and cloud resources
 - Supports Tier0, production and analysis **simultaneously**
- Looking towards **HL-LHC**
 - **Preserve the successful Global Pool model**
 - Build on **community** solutions leveraging **Dirac-X** transformation system
 - Design the next-generation Workflow Management **around GlideinWMS and HTCondor**



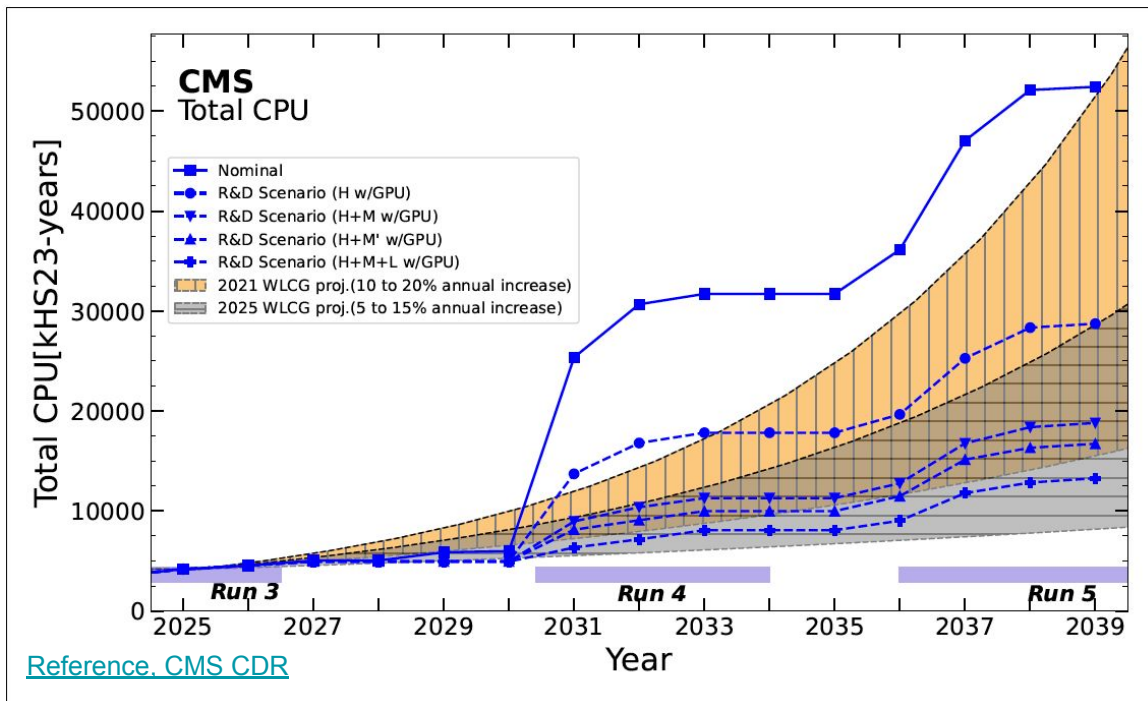
Additional slides



The HL-LHC Challenge for CMS Submission Infrastructure



The HL-LHC challenge for the CMS SI: continuous evolution enabling further **scalability** and **resource heterogeneity**, while maintaining the demonstrated stability, flexibility and **efficiency**



The SI vision for HL-LHC:

- Build from the current system that has served CMS effectively for more than a decade
- Expand and adapt to more and new resource types, diverse allocation policies and tools, etc



Sources of CPU Inefficiency

- Payload Inefficiencies
 - Bootstrapping and staging
 - I/O-bound jobs
 - Either **heavy I/O** jobs or jobs that use **remote reads**
 - User code (CRAB jobs)
 - StepChain (vs TaskChain): Multiple executables linked together as a single payload job
 - Pro: less jobs to manage, reduce intermediate data storage and transfers. 10x **faster turnaround**.
 - Con: diverse resource needs leading to **inefficiencies**
- Scheduling Inefficiencies
 - Non-standard requirements for jobs
 - System optimized for **2GB per core of RAM and 8 hours** of walltime
 - **Limited pilot lifetime**: draining and defragmentation

Valid reasons for inefficiencies, hard to reduce often.

- Scheduling efficiency typically >95% level for stable sites (T1s and big T2s)



CMS Connect



Analysis
(CRAB)

Tier 0

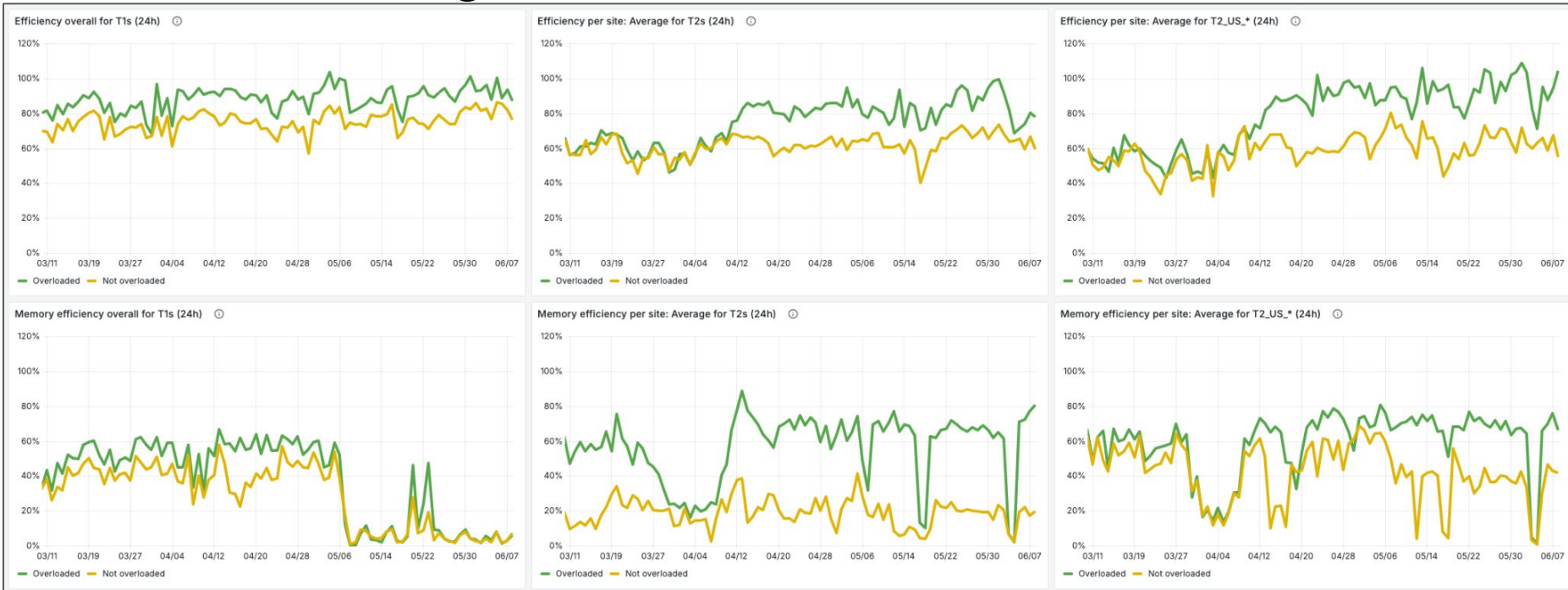
Production
(WMAgent)

CMS Connect

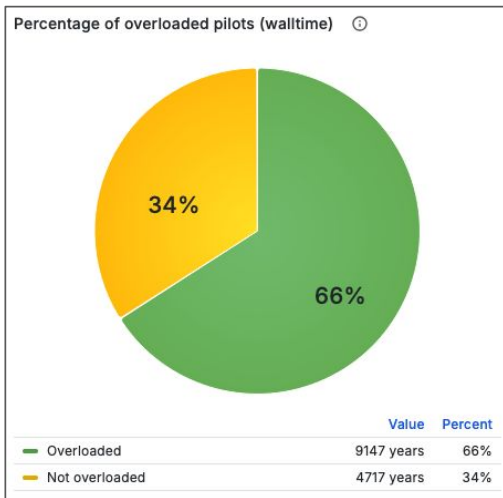
- Light interface for end user that allows to send plain condor jobs
- E.g.: Detector calibration studies not suitable for CRAB



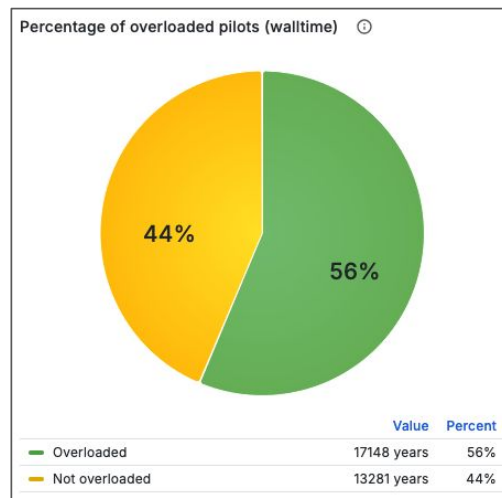
Overloading results



Overloading results



Percentage of overloaded pilots for T1s



Percentage of overloaded pilots for all sites